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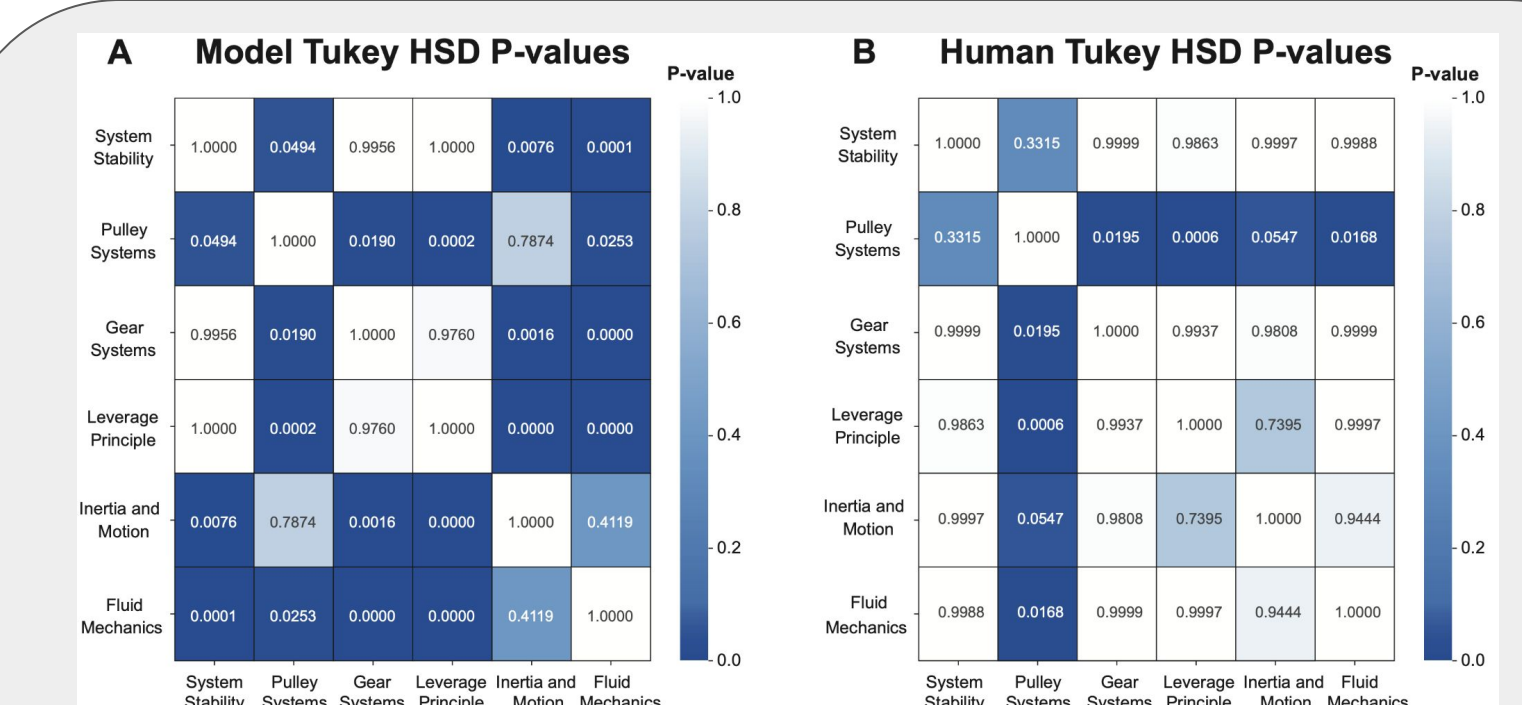
Mechanical reasoning is a key aspect of human intelligence. It involves understanding the interactions between physical objects, a skill that is key to humans' dexterity and problem-solving abilities. With the rapid development of visual-language models (VLMs) that merge vision and language capabilities, evaluating their mechanical reasoning capabilities is crucial to understanding their limitations. Underlying this ability is mental simulation, where individuals reason about mechanical systems by creating a mental model of the world.

## Cognitive Experiments

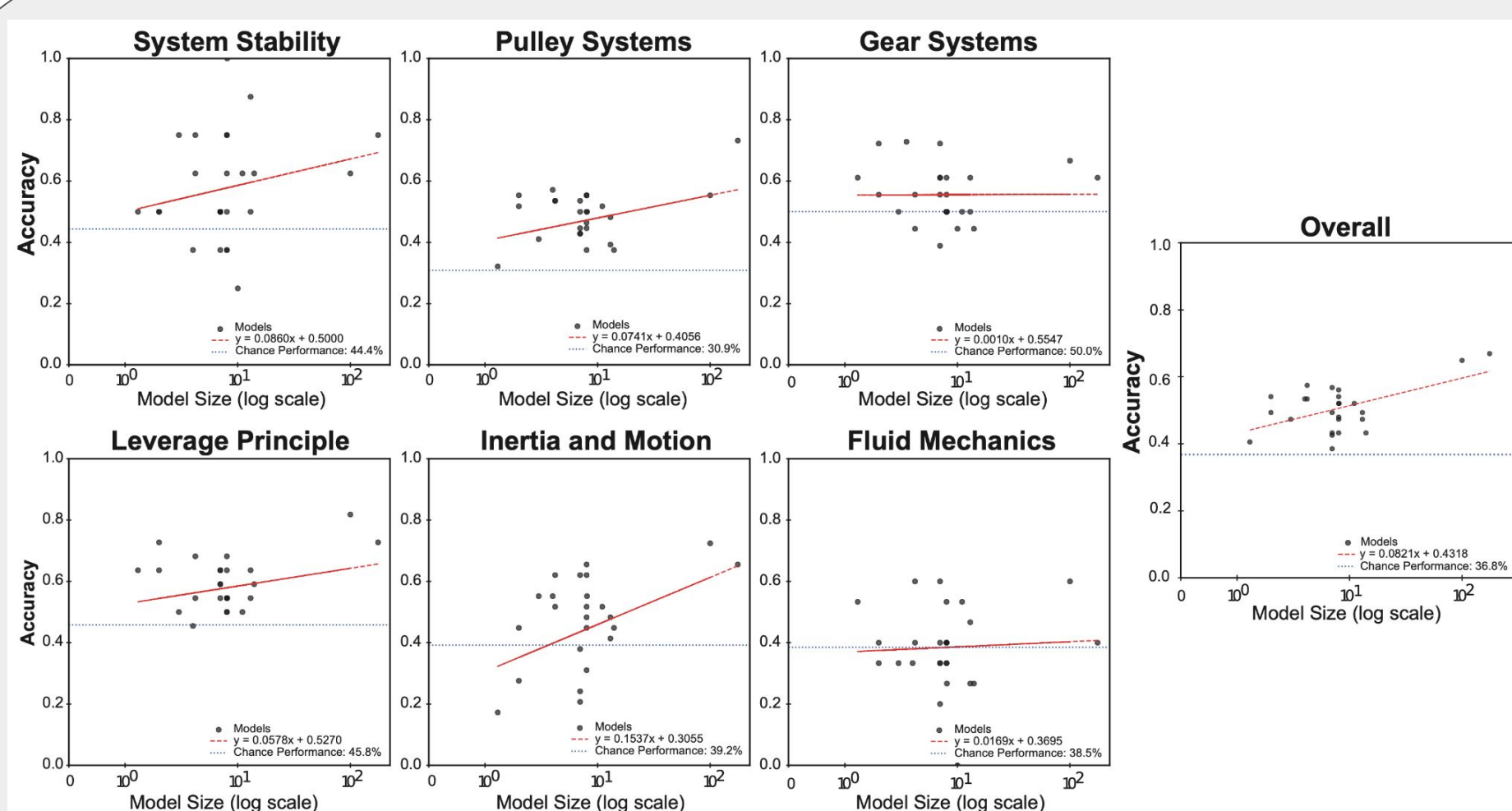
1. **System Stability:** Assesses the ability to predict the stability of physical systems.
2. **Pulley Systems:** Evaluates how VLMs reason about force distribution and motion in pulley setups.
3. **Gear Systems:** Tests logical and causal reasoning with gear mechanisms.
4. **Leverage Principle:** Involves understanding the relationship between force, distance, and torque
5. **Inertia and Motion:** Focuses on predicting object motion and force impacts over time.
6. **Fluid Mechanics:** Examines reasoning about fluid behaviors, considering factors like flow, pressure, and volume changes in dynamic scenarios.

## Main Results on MechBench

Among the evaluating dimensions, Pulley Systems exhibits the largest performance gap, with human accuracy nearing 90%, compared to model accuracy averaging around 50%.



- A. The ANOVA results for model performance reveal a significant difference across the six dimensions, indicating that the models exhibit varying accuracy depending on the task.
- B. Human performance across the six dimensions shows no significant difference.



In most of the dimensions, accuracy increases with larger model sizes. However, Gear Systems and Fluid Systems show very weak relationships.

But even they are far inferior to human results.

